

# TCS349: Responsible and Explainable AI

## Unit 3, Lecture 4

### AI, Inequality & the Concentration of Power

Who benefits most from AI-driven growth, why power concentrates among a small number of firms, and how unequal access widens the digital divide.

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**Economic Inequality**



**Concentration of Power**



**Digital Divide**



**Critical Perspectives**

## 1. AI and Economic Inequality - How the Gap Widens

AI's productivity gains are not distributed evenly - they tend to flow disproportionately to capital owners and highly skilled workers who can direct AI tools.

Workers whose tasks are most automatable often see wage stagnation or job loss, while those who complement AI see rising demand and pay.

This dynamic can widen the gap between high-skill and low-skill wages, even as overall economic output grows.

Existing inequality can compound over time, since capital gained from early AI adoption often funds further AI investment.

## 2. Winner-Take-Most Dynamics in AI Markets

Building competitive AI systems requires enormous data, computing infrastructure and talent, favouring firms that already have scale.

Once an AI product gains enough users, the resulting data can improve the product further, reinforcing the leader's advantage.

These dynamics make AI markets prone to a small number of dominant firms rather than many competing equally.

Smaller firms and startups increasingly depend on a handful of large firms for the cloud computing and AI infrastructure they need.

### 3. Concentration of Power Among AI Developers

A small number of companies control the most advanced AI models, the computing infrastructure to train them, and the platforms to distribute them.

This concentration gives these firms outsized influence over what AI capabilities exist and how they are governed.

Decisions made by a handful of organisations can shape norms, safety standards and access for the entire industry.

Concentrated power raises accountability questions similar to those in Unit II, but now at an industry-wide, rather than single-system, scale.

## 4. The Digital Divide - Access to AI's Benefits

The digital divide refers to unequal access to the internet, devices and digital skills needed to benefit from AI technologies.

Without reliable connectivity or affordable devices, individuals and businesses cannot access AI tools even when they are available.

The divide exists within countries, between urban and rural areas, and between wealthier and poorer nations globally.

AI can widen an existing digital divide, since those already connected gain new capabilities faster than those who are not.

## 5. Geographic and Global Inequality in AI Adoption

Most cutting-edge AI research, investment and infrastructure is concentrated in a small number of countries and regions.

Nations without this infrastructure risk becoming primarily consumers of AI products built elsewhere, rather than shapers of the technology.

Language and cultural bias in training data can make AI tools work less well for regions underrepresented in that data.

Building local AI capacity is increasingly treated as a matter of economic competitiveness and national policy, not just technical development.

## 6. Labour Market Polarisation

AI adoption has been associated with growth at both the high-skill and low-skill ends of the labour market, while many routine middle-skill roles shrink.

High-skill roles benefit from AI as a productivity multiplier, and low-skill roles often remain hard to automate due to physical or interpersonal demands.

Middle-skill, routine-task roles - many clerical and administrative positions - are the most exposed to this polarisation.

This pattern connects directly to the task automation and displacement dynamics covered in Lecture 1 of this unit.

## 7. Critical Perspectives - Competing Views on AI and Inequality

Optimistic view: AI is a general-purpose technology that will eventually raise living standards broadly, as previous technological revolutions did over time.

Critical view: without deliberate policy intervention, AI's gains will concentrate further among those who already own capital and data.

A middle position argues the outcome is not predetermined - it depends on choices made now about taxation, education and regulation.

This debate underpins the governance and public-policy questions examined later in this unit.